NYC Taxi Analytics

The Challenge

You're given **NYC Yellow Taxi trip data** — millions of real rides with pickups, dropoffs, fares, tips, timestamps, and zones.

Your mission:

> 🎯 **If you were advising a new taxi driver on how to maximize earnings, what would you tell them?**
> Think: best zones, best hours, tipping patterns, short vs. long trips, airport runs, payment method effects — surprise us with angles we haven't thought of.

There is no single right answer. We want to see **how you think**, what questions you ask of the data, and how you build a pipeline that could run reliably in production.

> 📐 **Design for scale.** You're working with one month of data, but assume this pipeline will eventually process **5 years of monthly data and run daily**. Design your pipeline and data model accordingly.

Dataset (Public — Ready to Use)

Yellow Taxi Trip Data (Parquet)

LINK TO DOWNLOAD

```
https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2023-01.parquet
```

Taxi Zone Lookup (CSV)

```
https://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv
```

What We Want

1. Use GCP — find the most cost-effective path

You must use **GCP**. You have freedom to choose whichever GCP services you think are best for this problem — but we're watching **how you optimize for cost**.

- BigQuery free tier (1 TB query/month, 10 GB storage) is available
- You can use any GCP services (Cloud Functions, Dataflow, Dataproc, etc.)
- **Justify your choices** — why this service over the alternatives? What does it cost?
- **Provide a rough monthly GCP cost estimate** for your pipeline running in production (5 years of data, daily runs)

2. Build a robust pipeline

Build a pipeline that:

- **Ingests** the source data
- **Cleans & enriches**
- **Produces analytical queries** that answer the driver earnings question

The pipeline should be **incremental** — not a one-time script:

- It should be able to process **new months** as they arrive without reprocessing old data

3. Deliver the queries

The core deliverable is a set of **analytical queries** that produce insights on maximizing driver earnings.

Go beyond the obvious. We want to see creative angles — the kind of analysis that makes someone say *"huh, I didn't think of that."*

4. Document your decisions

A short `README.md` covering:

1. **GCP services used** — what you picked and why (cost reasoning)
2. **Cost estimate** — rough monthly GCP cost for production (5 years of data, daily runs)
3. **Data model diagram** — show your staging/transformation layers (bronze → silver → gold, or your own approach)
4. **Pipeline design** — how it works, how to run it, what makes it robust
5. **Data quality issues** — what you found and how you handled them
6. **Your findings** — the actual advice to the driver, backed by your queries

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Rules

- **Time limit:** ~2 hours
- **Must use GCP** — pick the most cost-effective services and justify your choices

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Deliverables

- [ ] Pipeline code (ingest → clean → enrich)
- [ ] Data model diagram (layers and schema)
- [ ] Analytical queries with results
- [ ] `README.md` with decisions, cost estimate, findings, and reasoning

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